

Global Investment Research

Entering the Agentic AI Era for the Commerce Landscape

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Overview – Key Takeaways

Entering the Agentic AI Era for the Commerce Landscape

AI is reshaping the consumer shopping journey. This report frames how that shift unfolds, where value is migrating and who captures it, and where the debates are.



Why?

Implications span **eCommerce/retail, payments, digital advertising, and cybersecurity** — raising new debates on customer ownership, distribution economics, and merchant margins and liability.



When?

A long-duration transition of 3-5+ years, analogous to the development of eCommerce itself.



Where?

Adoption is currently concentrated in product discovery, research, and recommendations, though **we expect steady, measured progress** toward end-to-end agentic shopping.

Overview – Framing Four Implications

Our central conclusion: value should increasingly follow consumer intent

1

eCommerce. Expect agentic commerce to drive gradual increases to eCommerce penetration rates. **We estimate \$2.6tn of U.S. spend sits in high-likelihood categories (i.e., low-risk, recurring purchase types)** and every 2% of agentic penetration of card-present spend could add ~1% to eCommerce growth.

2

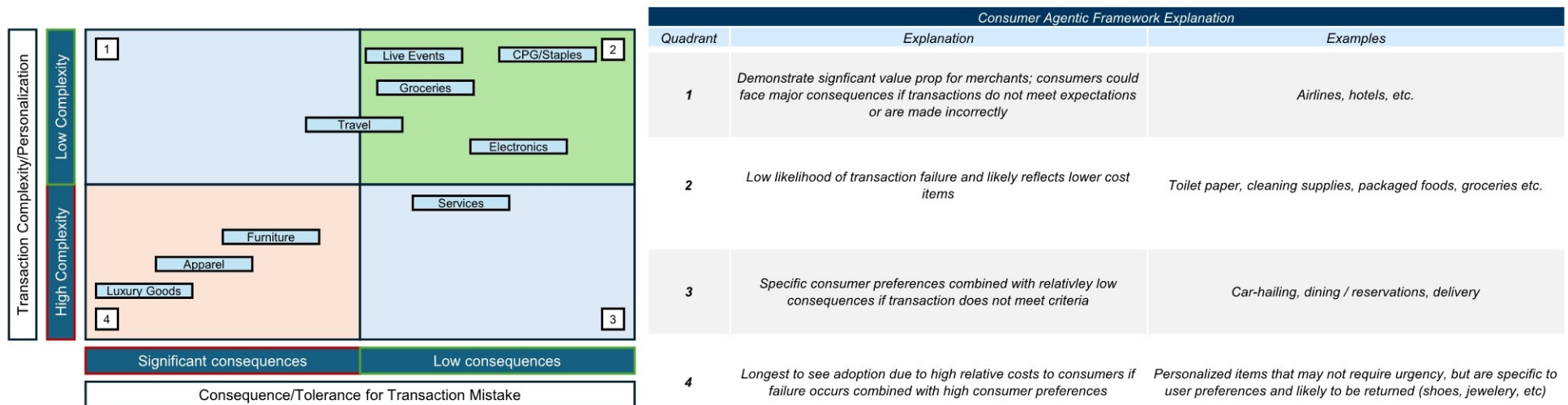
Digital Advertising. Parallel to the desktop-to-mobile shift, **expect ad monetization to migrate with commercial intent rather than disappear** as discovery moves inside AI interfaces and dollars flow to AI-native sponsored recommendations.

3

Payments. Agentic payments **build off existing card infrastructure**: protocols from Google, Stripe, Visa and Mastercard are wiring agent-to-merchant authorization, leaving card networks (V/MA) well positioned as liability frameworks mature.

4

Privacy & Security. The **threat environment may scale faster than legitimate autonomous commerce** and drive an increasing amount of demand for verified agent identify and fraud infrastructure.



Source: Goldman Sachs Global Investment Research

Key Investor Debates Across Agentic Commerce

Market Intelligence

Where we have seen the most debate...

Who Owns the Consumer?

- Large retailers may resist ceding discovery and the customer relationship to horizontal agents due to risks of weakening first-party data, loyalty, cross-selling, and retail-media economics

Does Agentic Commerce Democratize Discovery or Reinforce Scale

- Agents could improve discovery of smaller merchants (SHOP)
- Optimization around price, availability, delivery speed, and transaction reliability could simultaneously favor scaled players AMZN & WMT

Does Search Disruption Simply Create a New Toll Collector?

- AI agents could initially reduce dependence on paid search, but scaled agent platforms may ultimately monetize their control of consumer intent through various methods
- Likely that evolution of the top/middle of the advertising funnel will be a critical industry development

Can Trust Infrastructure Keep Pace with Product Innovation?

- The technology enabling agentic transactions is advancing rapidly, but widespread adoption depends on clear rules around intent verification, erroneous purchases, fraud losses, chargebacks, and returns

Beneficiaries

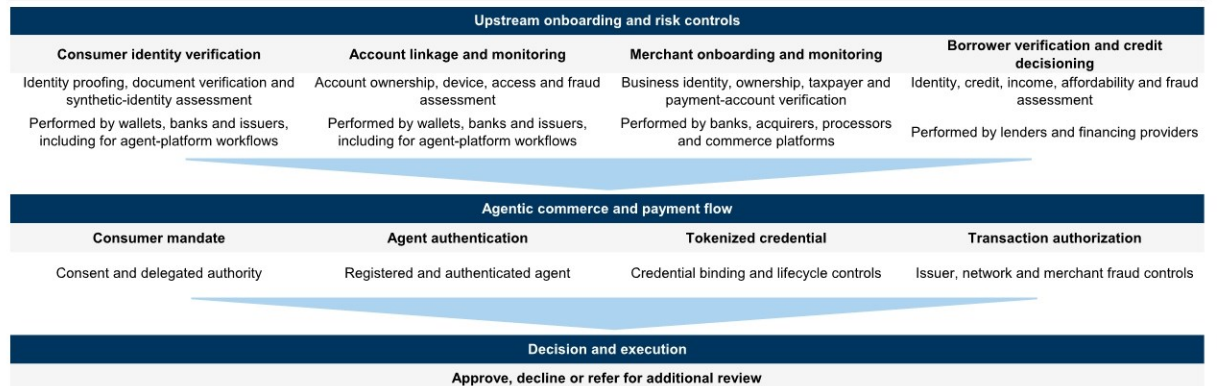
- **We believe winners will be determined less by raw AI capability**
- What qualities do we think win?
 - Distribution
 - Consumer Trust
 - Merchant Participation
 - Transaction Infrastructure
 - Ability to Capture Commercial Intent
- What sectors are beneficiaries?
 - eComm & Comm Infra
 - Payments
 - Identity
 - Fraud Prevention
 - Cybersecurity

Agentic Commerce Expands the Need for Identity, Fraud and Decisioning Infrastructure

Key Takeaways:

- Agentic commerce adds new identity and risk decisions around the consumer, merchant and borrower, complementing the authentication, tokenization and authorization controls provided by payment infrastructure.
- Legitimate agents may weaken traditional fraud signals as activity increasingly originates from shared cloud infrastructure, follows programmatic navigation patterns and occurs at machine speed.
- Consumer verification should remain risk-based, with additional checks concentrated around new account linkages, material identity changes, unfamiliar devices and broader delegated authority.
- Agent-led discovery could increase the importance of merchant monitoring, particularly for unfamiliar counterparties, mismatched digital storefronts and changes in merchant ownership or payment accounts.
- Agent-led financing could move eligibility and pricing decisions earlier in the purchase journey, increasing demand for borrower data and real-time, governed decisioning infrastructure.
- Payments fraud remains a key adoption risk, with dispute abuse and card testing cited more frequently than refund abuse, instant-payments fraud and phishing-related attacks.
- Agentic commerce could accelerate cybersecurity investment as legitimate automation becomes harder to distinguish from malicious activity, increasing demand for AI and pre-authorization detection.

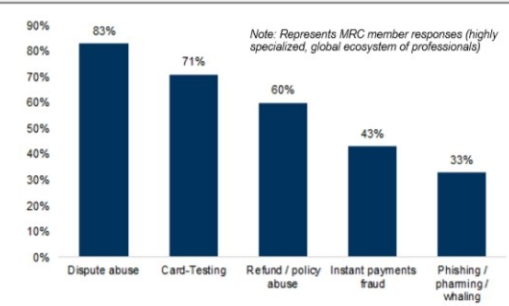
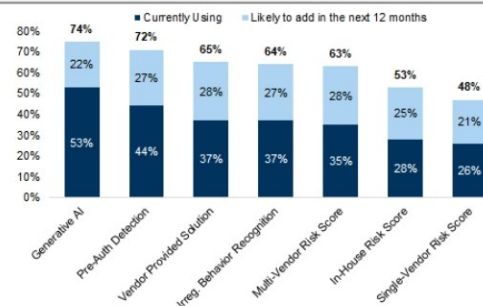
Agentic commerce adds identity and risk inputs around existing payment controls



Traditional fraud signals may become less reliable as agents operate through shared infrastructure and at machine speed

Fraud Signal	Status under Agentic Commerce	Rationale	Complementary Signal
Device and Network Signals			
Device reputation	Potentially misleading	Origination device may be cloud infrastructure shared across unrelated users	Verified agent credential and consumer-agent linkage
Devices linked across accounts	Potentially misleading	Shared model-provider infrastructure may connect unrelated consumers	Persistent consumer, account and device relationships
Proxy, VPN and hosting detection	Potentially misleading	Hosting infrastructure may be the normal execution path for authorized agents	Agent registration and authorization context
IP geolocation	Potentially misleading	IP address may belong to the model provider and be shared across users	Consumer, account and transaction context
Cookie and session persistence	Less available	Agent sessions may be ephemeral and lack conventional returning-customer history	Persistent identity and account relationships
Behavioral Signals			
Typing cadence and cursor movement	Unavailable	No direct human interaction exists at the point of execution	Agent credentials and consumer authentication
Navigation patterns	Potentially misleading	Navigation is programmatic and more uniform than human browsing	Mandate, merchant and transaction context
Session velocity	Potentially misleading	Authorized transaction activity may exceed normal human speed	Real-time transaction anomaly detection

Demand for fraud prevention is broad-based and increasing as merchants face growing fraud risks



Stakeholder Implications

Digital Advertising | Value Migrates with Commercial Intent

Our view: Advertising historically follows commercial intent. Near term, AI mainly enhances efficiency (creative cost, optimization, ROAS); over time a greater share of commercial intent originates inside AI interfaces rather than search or retailer sites.

Value Migration: Desktop-to-Mobile Parallel



Stock Highlights | Positively Skewed to the Shift in Commercial Intent

Digital Advertising

META

Personal AI & ads inside AI-native surfaces

Leading consumer-facing personal agent offerings with monetization paths across both commerce and advertising.

GOOGL

Owns discovery

AI interfaces extend discovery beyond traditional search, while emerging agent/payment protocols position it in the infrastructure converting intent into transactions.

eCommerce

AMZN

Intent closest to the transaction

Retail media sits where purchase intent already resides; offer competitive prices, availability and delivery reliability - what consumers (and increasingly agents will) optimize for.

Retailers & Marketplaces | Hardlines: Walmart

Agentic shopping tools for customer exploration

Sparky: Walmart's commerce agent

- Launched in June 2025
- Capabilities include customer service, building a cart with the same items a customer frequently purchases, synthesizing reviews/offering recommendations, and helping customers plan, compare, and purchase items
- Sparky has seen rapid adoption (weekly active customers doubled y/y, with 60% growth q/q) and drives a 40% higher AOV
- Customers are using Sparky to solve complex, multi-item shopping journeys (e.g., party planning)
- We expect a notable change in categories with room for inspiration that benefit from conversation (e.g., fashion, baby, beauty, auto care, and electronics)

Partnerships with GenAI platforms

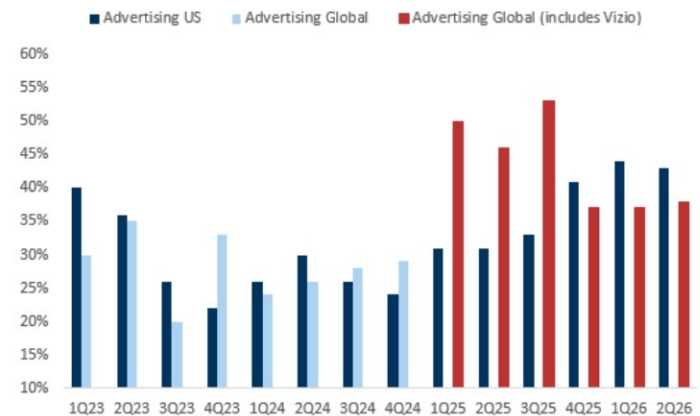
- Potential growth opportunity: 1) positions Walmart well to serve customers with its combination of assortment, price, and speed, 2) a Walmart-powered experience shows up within these agents, and Walmart receives customers who are in the midst of placing an order, and 3) Walmart can serve customers for occasions that do not begin with commercial intent
- Walmart could see a sales lift from these partnerships given large user bases (potential new customers) and ability to serve customers on new occasions
- We expect to see increased personalization over time, which could lead to incremental sales from existing customers

Four primary use cases for agent-based shopping: replenishment, mission-based, spearfishing, and inspiration/discovery

How will AI impact Walmart's advertising business?

- Sparky: For Walmart's hyper-users, the ad load is less, while the ad value is higher, and AI is a driver of ad growth (i.e., rather than a search query that leads to placement, contextual signals lead to activation, and the ad shows up as a solution versus just an advertisement)
- Walmart is partnering with AI players to capture top-of-funnel discovery and customer acquisition; third party agents are often programmed to optimize for retail metrics such as lowest price, fastest speed, and broadest assortment, and Walmart often wins on these metrics and emerges as the recommended solution
- Walmart maintains strict boundaries, sharing only perishable data (i.e., real time pricing and availability) while keeping long term customer data secure
- Omnichannel benefits: Walmart's large store network acts as fulfillment nodes, allowing for quick delivery; digitally engaged customer cohorts (such as Gen Z) often still prefer to complete purchases in store

Advertising growth at Walmart, y/y



Source: Company data, Goldman Sachs Global Investment Research

Retailers & Marketplaces | Consumer Staples

Potential Impact | Bifurcated by category involvement

Consumer involvement rises — impact shifts from disruption toward resilience

Commoditized / low-involvement — more tangible disruption

- Agents can own the full purchase cycle — paper goods, cleaners, center-of-store food, apparel staples
- On out-of-stocks, agents swap SKU or brand; private label wins on value
- Reorders run off prior purchase history with minimal human involvement
- Lower brand loyalty means value and availability drive the agent's choice
- Consumer involvement likely falls further as agents take over routine replenishment

At risk

CLX, KHC — private label exposure;
ODD — reliance on first-party data

Personalized / high-involvement — relatively more resilient

- Agents recommend, consumers still decide — e.g., prestige beauty and skincare
- Higher price points and efficacy claims sustain brand preference over private label
- Brand attributes must answer agent queries, e.g., best face cream for dry skin
- Loyalty and DTC investment keep consumers engaged in the final decision
- Consumers can override agent recommendations, preserving the payoff from brand building

Well positioned

EL — Partnerships with Shopify for DTC experience; Profound for AI-native brand discovery

Fewer slots than page 1 of search: agents surface only a handful of options, making machine-readable brand attributes paramount

Key Focus Areas | How brands optimize for agentic AI



From SEO to GEO

Ensure AI systems can ingest, understand and recommend your products



Machine-readable content

Attributes on-site, not buried in PDFs. MDLZ: Oreos in only 10% of AI cookie recs; PG pursuing GEO



Shift the marketing mix

Budgets tilt toward personalized categories where consumers still decide



Scale AI across operations

Campaigns, innovation and supply chain (PG, HSY)

Retailers & Marketplaces | Apparel & Softlines

Factors to Consider for Softlines Retail

Brand Differentiation Matters More

- We believe agentic commerce will increase the importance of brand differentiation and storytelling
- We expect marketing focus and in-person experiences to be increasingly important in softlines

Bifurcating outcomes based on consideration

- We expect bifurcation in outcomes based on commoditized vs. considered purchases; debates will remain high in basics and price-driven retail
- Product innovation, differentiated features, and recommendation-led discovery can influence the purchase decision

Optimizing for AI Discovery

- As shopping shifts to agentic, brands must make product attributes and differentiation easy for AI engines to understand and recommend
- This should favor technology-forward brands that pair agentic optimization with continued investment in brand-building and customer engagement

What Wins in Agentic Discovery

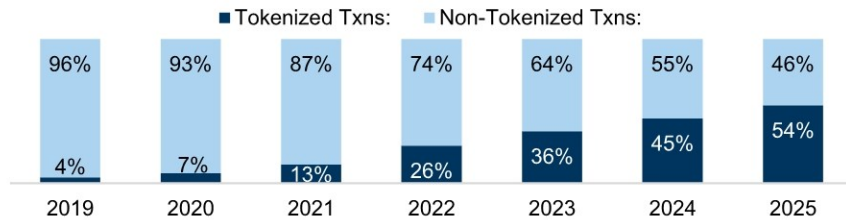
- We believe winners will be brands that AI agents can easily understand and recommend
- What brand attributes win?
 - Product Innovation
 - Differentiated Features
 - Brand Storytelling
 - AI Engine Optimization
 - Top-of-Mind Awareness
- Which categories screen best?
 - Considered Purchases
 - Product Innovation-Led
 - Higher Involvement
 - Recommendation-Led
 - Repeat Engagement

Stock Recommendations: SharkNinja (SN) & Tapestry (TPR)

- SN participates in more considered, innovation-led categories with lower substitution risk as AI narrows the recommendation set
- TPR's brand equity and customer engagement support preference in categories where agents can easily surface substitutes
- Both have been technology-forward, positioning them ahead of the average brand in agentic discovery

Payments & Financial Technology | Constructive on Networks and Modern Credit Issuers

V has shown the ability to shift away from manual entry checkout to tokenized txns



Note: Reflects CNP txns, including both "bottom" checkout and card-on-file

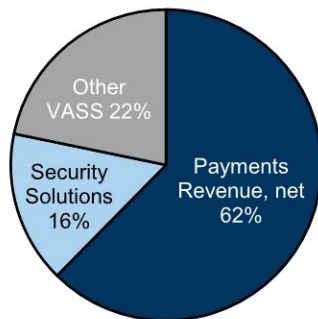
Where fraud liability falls within the payments ecosystem still needs to be figured out

Fraud Liability by Transaction Type				
Typical party responsible for fraud losses				
Transaction Type	Card-Present Transaction	Card-Not-Present Transaction	Non-Card Payments (varies)	Agentic Commerce
Responsible Party	Issuing Bank	Merchant	Consumer	TBD

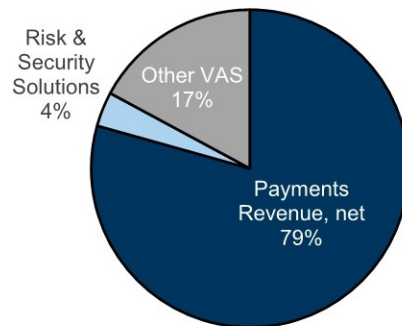
Fraud Risk Depends on the Several Key Factors of the Transaction

VAS, including cyber-related, act as asymmetrical growth engines for its businesses

MA Revenues, net (2024)



V Revenues, net (2024)



We view agentic commerce as a potential catalyst to accelerate eComm

1) Card Networks potentially benefitting from:

- Faster eCommerce growth
- Yield enhancement from higher value-added services (VAS) attach
- Yield enhancement from consumer basket fragmentation as a result of the distribution of agentic volumes across smaller merchant

2) Agentic commerce raise the technical bar for payment processing

- ## 3) Modern consumer finance (BNPL) issuers benefitting from consumers choosing:
- More Flexible
Cheaper
Simpler

Protocols, Network Risk, and Cybersecurity Product Portfolios

- We view existing payment ecosystem players as best equipped to provide leading infrastructure into a shifting consumer behavior environment
- We believe the scale of the networks positions V and MA well to sell products in these markets and provides an optimal solution to their acquiring and issuing clients, creating a suite of sticky mission critical products
 - We estimate that a 5% acceleration of V / MA's cybersecurity related value-added services would drive a 90bps/ 200bps acceleration VAS growth and 20bps / 80bps acceleration in total net revenue respectively

Business & Info Services | Positive on Identity, Fraud and Credit Decisioning Solutions

Key Takeaways:

- Agentic commerce could create verification and decisioning opportunities across four areas: consumer identity, merchant legitimacy, transaction fraud and financing eligibility.
- EFX has the broadest verification coverage, combining consumer identity and synthetic-risk tools with business, principal, taxpayer, payment-account, credit, income and employment data across the transaction lifecycle.
- TRU is differentiated in persistent identity resolution and network intelligence, linking consumers with accounts, devices and digital activity while detecting connected risks across transactions in real time and at scale.
- FICO is primarily positioned at the decisioning layer, with Scores supporting borrower-risk assessment and its Software portfolio supporting credit optimization, transaction-fraud detection and enterprise fraud decisions.

EFX, TRU and FICO address different layers of agentic identity, fraud and credit decisions

Potential decision	TRU	EFX	FICO	Potential buyer
Is the consumer legitimate?	Identity graph, document, device and IP intelligence	Identity proofing, synthetic identity and Kount digital-risk signals	Fraud decisioning using client and third-party inputs	Issuer, wallet, agent platform or lender
Is the merchant legitimate?	Device, IP and relationship analysis across storefronts	Business, principal, taxpayer and payment-account verification	Limited direct exposure	Acquirer, network, platform or agent operator
Is the transaction anomalous?	Persistent identity and network context	Consumer, account and digital-risk signals	Falcon and Enterprise Fraud decisioning	Issuer, processor or financial institution
Should financing be offered?	Credit, affordability, identity and fraud data	Credit, income, employment, identity and affordability data	Scores, Platform, rules, pricing and governance	Lender or financing provider

Existing EFX, TRU and FICO capabilities could extend into new agentic verification and decisioning workflows

Company and capability	Existing function	Potential agentic application
TransUnion		
TRU Identity Verification	Verifies consumer-provided information using credit, non-credit, public and digital identity data	Assess the consumer behind an agent credential when higher-risk changes occur
TRU Identity Graph / Neustar OneID	Connects physical and digital identifiers into a persistent identity view	Link consumers with accounts, devices and digital activity surrounding agent-led transactions
TRU Document Verification	Authenticates government-issued identification	Support step-up verification for higher-risk changes or financing requests
TRU Synthetic Fraud Model / Fraud Alerts	Identifies synthetic, altered or high-risk identities	Detect synthetic-identity risk in agent provisioning, account linkage or financing
TRU Device Risk / IP Intelligence	Assesses device reputation, account linkages and network characteristics	Provide context when legitimate agents operate through shared cloud infrastructure
TRU Fraud Analytics	Identifies relationships across identities, devices and transactions	Detect connected risks across consumers, accounts and digital storefronts
Equifax		
EFX Kount 360	Combines identity, payment-fraud data, analytics and decision workflows	Support identity and digital-risk assessment around agent-led activity
EFX Identity Proofing / Digital Identity Trust	Assesses whether identity information belongs to the represented consumer	Evaluate the underlying consumer behind an authenticated agent
EFX Document Verification	Authenticates government-issued identification	Support selective step-up verification
EFX Synthetic Identity Risk	Identifies synthetic-identity risk across the customer lifecycle	Assess higher-risk account linkages and financing requests
EFX Digital Risk Screening	Uses device, email and digital-interaction signals	Identify account-opening, access and account-takeover risk
EFX Business / Account Verification	Validates businesses, principals, taxpayer information and commercial accounts	Assess unfamiliar merchants and match payment destinations to verified entities
Fair Isaac		
FICO Scores	Standardized measures of consumer credit risk used by lenders to support underwriting, pricing and account-management decisions	Support borrower-risk assessment when agents initiate financing comparisons or applications
FICO Platform	Enterprise platform combining data, predictive models, business rules, optimization and orchestration across credit and fraud decisions	Govern and execute a larger number of real-time fraud, eligibility, pricing and offer decisions
Credit Decisioning and Optimization	Automates eligibility, underwriting, pricing and offer decisions under lender-defined policies and constraints	Evaluate and optimize lender-approved financing alternatives generated through agent-led commerce
FICO Falcon Fraud Manager	Provides real-time fraud detection, behavioral profiling and case management across cards, account-to-account payments and other transaction channels	Identify anomalous automated payment activity using transaction, customer and channel context
FICO Enterprise Fraud Solution	Connects fraud intelligence across customer interactions, accounts, products and channels	Assess agent-led activity using a broader customer view instead of evaluating each transaction independently

Cybersecurity | Rising Need for Security Requirements

Key Takeaways:

- Agentic commerce increases security requirements around verifying agent identity, confirming consumer authorization and protecting merchants from malicious automated activity.
- NET is positioned at the merchant edge, using Web Bot Auth and Bot Management to distinguish legitimate shopping agents from malicious bots and enforce emerging Visa and Mastercard protocols.
- OKTA is positioned at the identity and authorization layer, linking agents to authenticated consumers and controlling the data, accounts and purchasing actions they are permitted to access.

NET and OKTA address complementary layers of agentic-commerce security

Potential decision	NET	OKTA	Potential buyer
Is the agent legitimate?	Web Bot Auth validates cryptographically signed agent traffic	Okta for AI Agents establishes agent identity, ownership and lifecycle	Merchant or agent operator
Is the agent authorized by the consumer?	Validates identity and intent signals carried through V and MA protocols	Auth0 links the agent to an authenticated user	Agent platform, merchant or payment provider
What can the agent access?	Bot Management, WAF and API Shield enforce storefront and API access	Token Vault and Fine-Grained Authorization restrict credentials	Merchant or agent platform
Should the action require approval?	Enforces merchant rules based on agent identity and stated intent	Async Authorization requests consumer approval for higher-risk purchases	Agent platform or merchant
Is the activity malicious?	Detects scraping, credentials, inventory hoarding, malicious requests	Monitors identity and access activity and can revoke compromised agent access	Merchant or agent operator

Existing NET and OKTA capabilities could extend into agentic workflows

Company and capability	Existing function	Potential agentic application
Cloudflare		
Cloudflare Web Bot Auth	Cryptographically authenticates automated traffic.	Verify registered shopping agents and distinguish browsing from purchasing intent.
Cloudflare Bot Management	Identifies and manages automated traffic.	Allow legitimate purchasing agents while blocking scrapers, fraudsters and inventory abuse.
Cloudflare API Shield	Discovers and protects application APIs.	Secure catalog, inventory, account and checkout APIs accessed directly by agents.
Cloudflare WAF	Protects websites and applications from malicious requests.	Prevent agents or malicious bots from exploiting merchant storefronts.
Cloudflare AI Crawl Control	Controls AI-crawler access to website content.	Determine which agents may access or pay for product and merchant data.
Okta		
Auth0 User Authentication	Combines identity, payment-fraud data, analytics and decision workflows	Support identity and digital-risk assessment around agent-led activity
Auth0 Token Vault	Assesses whether identity information belongs to the represented consumer	Evaluate the underlying consumer behind an authenticated agent
Auth0 Async Authorization	Authenticates government-issued identification	Support selective step-up verification
Auth0 Fine-Grained Authorization	Identifies synthetic-identity risk across the customer lifecycle	Assess higher-risk account linkages and financing requests
Okta for AI Agents	Uses device, email and digital-interaction signals	Identify account-opening, access and account-takeover risk

Entertainment | Opportunities & Risks to Live Event Ticketing

Our thesis: Agentic commerce has the potential to reshape discovery, purchasing behavior, and distribution economics across live-event ticketing, but the impact is likely to differ materially between primary and secondary ticketing platforms.

On net, we believe primary ticketing platforms are best positioned to benefit from the opportunities around agentic commerce due to their access to differentiated inventory, while secondary ticketing platforms could be more exposed to the risks due to their commoditized inventory.

OPPORTUNITIES (+)

Ease of discovery & personalization

- AI agents have the potential to reduce friction in the ticket-buying journey as they identify relevant events by price, location, artist and venue (instead of consumers navigating multiple websites).
- Ticketmaster & Meta Muse's integration show agentic discovery layered onto existing infrastructure; StubHub frames AI as an incremental acquisition channel.

Higher sell-through & utilization

- AI assistants can help surface events to consumers who are actively looking for things to do in a given city or time window.
- As a result, agents help ticketing platforms, promoters, and venue operators increase attendance and sell-through for events that have excess capacity.

PUTS & TAKES (+/-)

Customer acquisition costs

Customer acquisition costs represent a meaningful expense for many secondary operators due strong purchase intent and fairly commoditized inventory, whereas primary operators generally have less reliance on customer acquisition efforts given their more differentiated inventory.

Near term: margin tailwind

- If agentic channels can deliver consumers below today's paid-search cost, secondary marketplaces see CAC relief and margin expansion.

Long term: a new distribution toll

- As usage scales, agentic platforms likely gain pricing power, resulting in ticketing marketplaces paying referral, placement, and commission fees to secure visibility within AI-generated recommendations.

RISKS (-)

Commoditization of undifferentiated inventory

- All-in pricing plus frictionless AI-driven price comparison shifts differentiation from brand and marketing to price and fulfillment reliability.
- This dynamic could be particularly impactful in secondary ticketing, where inventory is often highly commoditized and the same seat type may be listed across multiple marketplaces, ultimately pressuring the platform's ability to sustain service fees.

Ancillary & advertising leakage

- Agents fulfill a defined request rather than browse for incremental offers, which could result in lower ancillary attach rates and pressure the value of paid placement on ticketing platforms.

Disclosure Appendix

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Disclosure Appendix

Reg AC

We, Eric Sheridan, Will Nance, George K. Tong, CFA, Kate McShane, CFA, Bonnie Herzog, Brooke Roach, CFA, Leah Jordan, CFA, Gabriela Borges, CFA and Stephen Laszcyk, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

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Quantum

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Disclosure Appendix

Disclosures

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The rating(s) for Adyen NV is/are relative to the other companies in its/their coverage universe: Adyen NV, Capgemini, Dassault Systemes, Indra, Nemetschek, Nexi, OvH Groupe, SAP, SECO, Sage Group, Sinch AB, TeamViewer, Temenos, TietoEVRY, Wise Group, Worldline

The rating(s) for Walmart Inc. is/are relative to the other companies in its/their coverage universe: Academy Sports & Outdoors Inc., Advance Auto Parts Inc., AutoZone Inc., BJ's Wholesale Club Holdings, Bath & Body Works Inc., Best Buy Co., Bob's Discount Furniture Inc., Boyd Group, Costco Wholesale, Dick's Sporting Goods, Dollar General Corp., Dollar Tree Stores Inc., Driven Brands Holdings, Five Below Inc., Floor & Decor Holdings, Genuine Parts Co., Home Depot Inc., Lithia Motors Inc., Lowe's Cos., National Vision Holdings, O'Reilly Automotive Inc., Ollie's Bargain Outlet Inc., Petco Health and Wellness Co., RH, Target Corp., Tractor Supply Co., Ulta Beauty Inc., Valvoline Inc., Walmart Inc., Williams-Sonoma Inc. Inc.

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The rating(s) for Mastercard Inc. and Visa Inc. is/are relative to the other companies in its/their coverage universe: Affirm Holdings, Bill.com Holdings, Block Inc., Chime Financial Inc., Fidelity National Information Services, Fiserv Inc., Flywire Corp., Global Payments Inc., Global-E Online Ltd., Jack Henry & Associates Inc., Klarna Group Plc., Marqeta Inc., Mastercard Inc., NCR Voyix Corp., PAR Technology Corp., PayPal Holdings, Paymentus Holdings, Remitly Global Inc., Riskified Ltd., Shift4 Payments, SoFi Technologies Inc., Toast Inc., Upstart Holdings, Visa Inc., Western Union Co.

The rating(s) for Shopify Inc. is/are relative to the other companies in its/their coverage universe: Adobe Inc., Akamai Technologies Inc., AvePoint, Braze Inc., Check Point, Cloudflare, Commerce.com Inc., CoreWeave Inc., CrowdStrike, Datadog Inc., DigitalOcean Holdings, EverCommerce Inc., Figma Inc., Fortinet, HubSpot Inc., Intuit Inc., Klaviyo, Microsoft Corp., Navan Inc., Octave Intelligence, Okta, Oracle Corp., Palantir Technologies, Palo Alto Networks, SailPoint Inc., Salesforce Inc., SentinelOne, ServiceNow Inc., Shopify Inc., Snowflake Inc., Twilio Inc., Tyler Technologies Inc., Veeva Systems Inc., Workday Inc., Zeta Global, Zscaler.

Company-specific regulatory disclosures

Compendium report: please see disclosures at <https://www.gs.com/research/hedge.html>. Disclosures applicable to the companies included in this compendium can be found in the latest relevant published research

Disclosure Appendix

Price target and rating history chart(s)

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Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	59%	43%

As of July 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,104 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Target price history table(s)**StubHub Holdings (STUB)**

Date of report	Target price (\$)	Closing price (\$)
14-May-26	16.00	8.55
05-Mar-26	15.00	8.91
14-Nov-25	45.00	14.87
13-Oct-25	46.00	19.61

SharkNinja, Inc. (SN)

Date of report	Target price (\$)	Closing price (\$)
06-Aug-26	204.00	179.93
29-Jul-26	165.00	159.22
12-Feb-26	144.00	129.02
04-Feb-26	135.00	116.18
07-Nov-25	129.00	92.44
20-Oct-25	127.00	92.81
08-Aug-25	132.00	116.06
21-Jul-25	126.00	114.81
09-May-25	112.00	92.25
22-Apr-25	100.00	74.47
14-Feb-25	133.00	110.31
01-Nov-24	116.00	89.12
21-Oct-24	123.00	109.86
09-Aug-24	102.00	87.64
10-May-24	83.00	71.84
25-Apr-24	74.00	62.79
16-Feb-24	59.00	52.51
25-Oct-23	52.00	40.68

Mastercard Inc. (MA)

Date of report	Target price (\$)	Closing price (\$)
30-Jul-26	701.00	577.35
30-Apr-26	674.00	502.92
13-Apr-26	713.00	508.58
29-Jan-26	739.00	543.73
06-Jan-26	710.00	580.34
30-Oct-25	713.00	553.68
13-Oct-25	665.00	559.24

Amazon.com Inc. (AMZN)

Date of report	Target price (\$)	Closing price (\$)
31-Jul-26	375.00	271.58
08-Jul-26	335.00	243.62
30-Apr-26	325.00	265.06
13-Apr-26	275.00	239.89
06-Feb-26	280.00	210.32
13-Jan-26	300.00	242.60
31-Oct-25	290.00	244.22

Mastercard Inc. (MA)

Date of report	Target price (\$)	Closing price (\$)
31-Jul-25	688.00	566.47
14-Jul-25	674.00	553.02
01-May-25	644.00	546.63
08-Apr-25	613.00	479.92
02-Apr-25	633.00	547.38
30-Jan-25	628.00	566.01
09-Jan-25	572.00	516.40
02-Dec-24	557.00	531.36
31-Oct-24	563.00	499.59
14-Oct-24	578.00	506.66
31-Jul-24	540.00	463.71
12-Jul-24	519.00	440.11
10-Jul-24	520.00	433.64
01-May-24	517.00	442.07
15-Apr-24	534.00	459.79
31-Jan-24	514.00	449.23
10-Jan-24	482.00	428.27
26-Oct-23	433.00	364.59

Amazon.com Inc. (AMZN)

Date of report	Target price (\$)	Closing price (\$)
03-Oct-25	275.00	219.51
01-Aug-25	240.00	214.75
21-Apr-25	220.00	167.32
07-Feb-25	255.00	229.15
01-Nov-24	240.00	197.93
02-Aug-24	230.00	167.90
15-Jul-24	250.00	192.72
01-May-24	225.00	179.00
02-Feb-24	220.00	171.81
10-Jan-24	200.00	153.73
27-Oct-23	190.00	127.74
12-Oct-23	175.00	132.33

Live Nation Entertainment Inc. (LYV)

Date of report	Target price (\$)	Closing price (\$)
31-Jul-26	210.00	174.13
08-Jul-26	202.00	182.14
06-May-26	192.00	167.82
11-Mar-26	190.00	165.83
20-Feb-26	175.00	162.67
01-Jan-26	165.00	142.50
08-Dec-25	160.00	138.08
05-Nov-25	162.00	134.79
02-Oct-25	170.00	156.67
08-Aug-25	168.00	153.13
17-Jun-25	162.00	145.59
02-May-25	157.00	133.77
17-Apr-25	155.00	127.22
21-Feb-25	170.00	149.40
13-Feb-25	166.00	152.58
12-Nov-24	148.00	129.67
24-Oct-24	132.00	116.45
03-May-24	128.00	94.66
23-Feb-24	125.00	95.32
03-Nov-23	122.00	85.36

Shopify Inc. (SHOP)

Date of report	Target price (\$)	Closing price (\$)
05-Aug-26	194.00	144.24
06-May-26	170.00	105.44
12-Feb-26	168.00	110.66
04-Nov-25	193.00	160.94
06-Aug-25	176.00	154.90
08-May-25	120.00	94.00
21-Apr-25	130.00	81.64
11-Feb-25	150.00	123.59
12-Nov-24	135.00	108.92
28-Oct-24	88.00	79.87
07-Aug-24	78.00	63.89
22-May-24	74.00	58.70
09-May-24	67.00	62.45
14-Feb-24	75.00	80.67
02-Nov-23	67.00	59.70

Equifax Inc. (EFX)

Date of report	Target price (\$)	Closing price (\$)
04-Sep-26	199.00	177.05
21-Jul-26	182.00	173.00
26-Jun-26	173.00	158.48
21-Apr-26	200.00	192.42
02-Apr-26	192.00	182.39
06-Feb-26	208.00	195.62
09-Jan-26	235.00	226.96
21-Oct-25	239.00	229.93
02-Oct-25	243.00	232.35
22-Jul-25	253.00	238.41
04-Apr-25	255.00	210.07
07-Feb-25	264.00	248.28
03-Jan-25	273.00	253.39
05-Dec-24	278.00	260.78

Alphabet Inc. (GOOGL)

Date of report	Target price (\$)	Closing price (\$)
23-Jul-26	435.00	317.69
08-Jul-26	440.00	361.92
30-Apr-26	450.00	384.80
05-Feb-26	400.00	331.25
12-Jan-26	375.00	331.86
30-Oct-25	330.00	281.48
14-Oct-25	288.00	245.45
24-Jul-25	234.00	192.17
10-Jul-25	225.00	177.62
25-Apr-25	220.00	161.96
11-Apr-25	205.00	157.14
05-Feb-25	220.00	191.33
13-Jan-25	215.00	191.01
30-Oct-24	210.00	174.46

Equifax Inc. (EFX)

Date of report	Target price (\$)	Closing price (\$)
17-Oct-24	293.00	280.30
02-Oct-24	312.00	293.33
20-Aug-24	307.00	296.58
18-Jul-24	277.00	264.19
31-May-24	246.00	231.39
19-Apr-24	241.00	216.20
01-Mar-24	276.00	270.98
08-Feb-24	252.00	252.06
26-Jan-24	241.00	246.12
29-Dec-23	222.00	247.29
01-Dec-23	195.00	222.12
30-Oct-23	173.00	164.02
19-Oct-23	188.00	178.98
29-Sep-23	208.00	183.18

Alphabet Inc. (GOOGL)

Date of report	Target price (\$)	Closing price (\$)
14-Oct-24	208.00	164.96
24-Jul-24	217.00	172.63
08-Jul-24	211.00	189.03
26-Apr-24	195.00	171.95
15-Apr-24	185.00	154.86
31-Jan-24	171.00	140.10
10-Jan-24	164.00	142.28
04-Oct-23	154.00	135.24

TransUnion (TRU)

Date of report	Target price (\$)	Closing price (\$)
04-Sep-26	89.00	79.88
28-Jul-26	82.00	83.80
26-Jun-26	72.00	70.35
29-Apr-26	75.00	70.47
02-Apr-26	74.00	69.32
13-Feb-26	80.00	73.92
05-Dec-25	92.00	84.47
23-Oct-25	86.00	83.48
03-Oct-25	78.00	77.34
02-Oct-25	83.00	73.51
29-Aug-25	96.00	88.40
24-Jul-25	102.00	98.43
25-Apr-25	90.00	82.09
04-Apr-25	87.00	71.10
28-Feb-25	102.00	92.43
14-Feb-25	107.00	99.74
03-Jan-25	101.00	92.27
05-Dec-24	111.00	98.42
23-Oct-24	118.00	108.67
02-Oct-24	109.00	105.90
20-Aug-24	97.00	93.30
26-Jul-24	87.00	86.46
01-Mar-24	81.00	78.71
14-Feb-24	78.00	77.53
29-Dec-23	72.00	68.71
30-Oct-23	63.00	43.30
25-Oct-23	73.00	46.23
29-Sep-23	85.00	71.79

Visa Inc. (V)

Date of report	Target price (\$)	Closing price (\$)
29-Jul-26	438.00	368.73
29-Apr-26	420.00	334.86
13-Apr-26	394.00	309.39
30-Jan-26	408.00	321.83
29-Oct-25	417.00	341.28
30-Jul-25	406.00	350.91
14-Jul-25	416.00	350.50
30-Apr-25	390.00	345.50
08-Apr-25	386.00	308.27
02-Apr-25	392.00	346.33
31-Jan-25	384.00	341.80
09-Jan-25	346.00	312.60
02-Dec-24	347.00	316.65
30-Oct-24	324.00	290.16
14-Oct-24	325.00	280.68
24-Jul-24	317.00	254.17
12-Jul-24	334.00	265.74
24-Apr-24	335.00	275.02
15-Apr-24	328.00	271.28
26-Jan-24	318.00	267.94
10-Jan-24	312.00	264.56
25-Oct-23	285.00	236.85
12-Oct-23	289.00	236.78

Fair Isaac Corp. (FICO)

Date of report	Target price (\$)	Closing price (\$)
30-Jul-26	1,548.00	1,139.54
26-Jun-26	1,424.00	1,182.79
29-Apr-26	1,543.00	1,043.57
02-Apr-26	1,528.00	1,089.76
02-Feb-26	1,770.00	1,450.91
06-Nov-25	2,070.00	1,674.80
03-Oct-25	2,074.00	1,850.65
31-Jul-25	1,915.00	1,436.72
30-Apr-25	2,244.00	1,989.68
04-Apr-25	2,144.00	1,673.98

Walmart Inc. (WMT)

Date of report	Target price (\$)	Closing price (\$)
08-Sep-26	132.00	106.05
21-Aug-26	130.00	103.70
20-Jul-26	141.00	112.20
21-May-26	154.00	121.34
20-Feb-26	138.00	122.99
21-Nov-25	121.00	105.32
21-Aug-25	114.00	97.96
16-May-25	101.00	98.24
16-Apr-25	99.00	91.19
21-Feb-25	106.00	94.78

Fair Isaac Corp. (FICO)

Date of report	Target price (\$)	Closing price (\$)
05-Feb-25	2,287.00	1,847.80
05-Dec-24	2,661.00	2,367.16
07-Nov-24	2,374.00	2,176.10
02-Oct-24	2,130.00	1,938.16
20-Aug-24	1,879.00	1,750.00
01-Aug-24	1,722.00	1,578.95
02-Jul-24	1,652.00	1,529.38
31-May-24	1,502.00	1,289.93
26-Apr-24	1,442.00	1,110.85
01-Mar-24	1,448.00	1,295.20
26-Jan-24	1,337.00	1,196.36
04-Dec-23	1,289.00	1,122.43
01-Dec-23	1,156.00	1,127.71
09-Nov-23	1,058.00	962.70

Walmart Inc. (WMT)

Date of report	Target price (\$)	Closing price (\$)
29-Jan-25	101.00	97.50
19-Nov-24	93.00	86.60
15-Aug-24	81.00	73.18
17-May-24	73.00	64.65
20-Feb-24	193.00	58.62
16-Nov-23	180.00	52.01

Adyen NV (ADYEN.AS)

Date of report	Target price (€)	Closing price (€)
14-Aug-26	1,850.00	1,061.60
24-Mar-26	1,800.00	888.50
12-Feb-26	1,900.00	903.30
30-Oct-25	2,225.00	1,516.40
01-Oct-25	2,100.00	1,368.00
15-Aug-25	2,200.00	1,396.40
02-May-25	2,300.00	1,468.00
11-Apr-25	2,200.00	1,330.60
24-Mar-25	2,300.00	1,507.00
17-Feb-25	2,400.00	1,841.20
30-Jan-25	2,200.00	1,563.00
11-Nov-24	2,100.00	1,303.20
19-Aug-24	1,950.00	1,313.60
13-May-24	1,900.00	1,258.00
13-Feb-24	2,000.00	1,444.20
11-Jan-24	1,600.00	1,164.20
24-Nov-23	1,425.00	1,073.60

Meta Platforms Inc. (META)

Date of report	Target price (\$)	Closing price (\$)
30-Jul-26	725.00	539.03
08-Jul-26	815.00	603.12
30-Apr-26	830.00	611.91
13-Apr-26	840.00	634.53
29-Jan-26	835.00	738.31
30-Oct-25	815.00	666.47
14-Oct-25	870.00	708.65
31-Jul-25	830.00	773.44
10-Jul-25	775.00	727.24
01-May-25	690.00	572.21
11-Apr-25	685.00	543.57
30-Jan-25	765.00	687.00
13-Jan-25	688.00	608.33
31-Oct-24	630.00	567.58
14-Oct-24	636.00	590.42
01-Aug-24	555.00	497.74
08-Jul-24	522.00	529.32
25-Apr-24	500.00	441.38
15-Apr-24	555.00	500.23
02-Feb-24	500.00	474.99
10-Jan-24	414.00	370.47

Tapestry Inc. (TPR)

Date of report	Target price (\$)	Closing price (\$)
28-Apr-26	169.00	143.84
06-Feb-26	166.00	152.10
29-Jan-26	146.00	127.33
07-Nov-25	129.00	103.68
03-Oct-25	130.00	114.92
14-Aug-25	116.00	95.69
21-Jul-25	118.00	106.41
09-May-25	93.00	78.01
22-Apr-25	84.00	64.92
07-Feb-25	98.00	80.38
10-Dec-24	72.00	61.39
14-Nov-24	66.00	57.82
08-Nov-24	58.00	52.22
21-Oct-24	52.00	44.51
09-Feb-24	47.00	42.00
10-Nov-23	41.00	28.12

Disclosure Appendix

Regulatory disclosures

Disclosures required by United States laws and regulations

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